Fuzzy Adaptive Manifold For Sigma Point Generation In a Square-Root Unscented Kalman Filter

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Abstract

This work introduces the Fuzzy Adaptive Manifold (FAM) for the Square-Root Unscented Kalman Filter, optimized for Cortex-M4 embedded systems. By implementing a Sigma Point Fuzzy Inference (SpFI) mechanism, the filter transitions from global covariance scaling to point-specific trust mapping. This allows for iteration-specific manifold warping, providing robust outlier rejection and directional sensitivity during sensor saturation. The implementation utilizes a Euclidean Proxy for real-time performance, achieving superior dynamic response in 800 Hz navigation loops.

1 Introduction

The Unscented Kalman Filter (UKF) and its numerically stable counterpart, the Square-Root Unscented Kalman Filter (SR-UKF), represent the state-of-the-art for non-linear estimation in embedded environments. By leveraging a deterministic set of sigma points to propagate the mean and covariance, the SR-UKF captures higher-order moments of the state distribution more accurately than the First-Order Taylor series approximations used in the Extended Kalman Filter (EKF). Despite these advantages, the standard Unscented Transform (UT) is fundamentally constrained by a rigid, symmetric geometry governed by global scaling parameters (α, β, κ).

In high-performance embedded applications, such as Inertial Navigation Systems (INS) deployed on Cortex-M4 hardware like the STM32F401, systems frequently encounter non-Gaussian noise, directional sensor saturation, and transient outliers. Traditional adaptive techniques—including our previous work on fuzzy-scaled process noise (Q_s)—rely on global inflation factors. While effective for maneuvering targets, these methods apply uniform scaling to the entire uncertainty ellipsoid. Such *vague* adaptation often results in over-correction, where the filter loses precision in reliable state dimensions to compensate for a single corrupted measurement axis.

This paper introduces **Fuzzy Adaptive Manifold (FAM)**, a novel framework that shifts the adaptive mechanism from global scaling to **Per-Point Fuzzy Importance Mapping**. Central to this architecture is the **Sigma Point Fuzzy Inference (SpFI)**, which evaluates the individual inferences of each trajectory in the state-space. By treating the uncertainty manifold as a deformable "fuzzy" manifold rather than a static ellipsoid, FAM allows the filter to selectively de-weight specific sigma points that correspond to non-physical maneuvers or localized sensor failures.

By warping the sigma point distribution in real-time, SpFI achieves the directional robustness typically associated with high-overhead Particle Filters, while maintaining the $O(L^3)$ computational efficiency required for deterministic execution on microcontrollers. This paper details the mathematical formulation of the SpFI mapping, its implementation via innovation whitening, and its integration into the Square-Root Unscented Kalman Filter architecture.

2 Fuzzy Adaptive Manifold (FAM)

The foundational principle of Fuzzy Adaptive Manifolds (FAM) is the transformation of the rigid covariance ellipsoid into a dynamic, deformable *trust manifold*. In a standard SR-UKF, the sigma points are dependent on the Cholesky factor S. Under FAM, we decouple the geometric location of the sigma point from its statistical influence, allowing the filter to project the uncertainty onto a new, non-linear manifold at every iteration.

2.1 Canonical Sigma Point Generation

In a standard SR-UKF, a set of $2L + 1$ sigma points $\mathcal{X}_i$ is generated as a function of the Square-Root Covariance S. For a state dimension L, the points are distributed as:

$$\mathcal{X}_0 = \hat{x} \tag{1}$$

$$\mathcal{X}_i = \hat{x} + \gamma S_i, \quad i = 1, \ldots, L \tag{2}$$

$$\mathcal{X}_{i+L} = \hat{x} - \gamma S_i, \quad i = 1, \ldots, L \tag{3}$$

where $\gamma = \sqrt{L + \lambda}$ is the global scaling factor. While these points define the geometry of the uncertainty ellipsoid, FAM introduces a mapping function $\Phi(\mathcal{X}_i)$ that evaluates the local innovation of each point, allowing for an asymmetric deformation of the uncertainty manifold during the update phase.

2.2 Sigma Point Fuzzy Inference (SpFI) Mapping

The SpFI mechanism replaces the static weight distribution of the Unscented Transform with a dynamic, point-specific importance mapping. Rather than adjusting the global spread α, SpFI evaluates the likelihood of each individual sigma point trajectory. We define a non-linear mapping $\mathcal{M} : \mathbb{R}^L \to [\phi_{\min}, 1]$ through a two-stage process:

1. **Whitened Innovation Mapping:** For each propagated sigma point, we generate a predicted measurement $\mathcal{Y}_{i,k} = h(\mathcal{X}_{i,k|k-1})$. The point-specific residual is mapped to a squared distance ζ_i using the triangular Cholesky factor $S_{r,k}$ via forward-substitution:

$$\zeta_i = \|S_{r,k}^{-1}(\boldsymbol{z}_k - \mathcal{Y}_{i,k})\|^2 \tag{4}$$

   where $\boldsymbol{z}_k$ is the current measurement vector and $\mathcal{Y}_{i,k}$ is the point-wise prediction.

2. **Fuzzy Inference System:** The scalar ζ_i serves as the input to the SpFI system. It is mapped through a fuzzy membership function $\mu_{\text{trust}}(\zeta_i)$ to produce the final importance factor ϕ_i:

$$\phi_i = \max\left(\phi_{\min}, \text{FIS}(\zeta_i)\right) \tag{5}$$

   where $\text{FIS}(\cdot)$ denotes a Zero-Order Sugeno inference engine that maps the whitened innovation ζ_i to a trust-weighting manifold.

Note: In this implementation, the Fuzzy Adaptive Manifold (FAM) is realized through a discrete sampling of the state-space using $2L + 1$ sigma points. The SpFI mapping ϕ_i acts as a local metric tensor, deforming the uncertainty surface at each sampled coordinate to reflect the real-time sensor consistency.

2.3 Asymmetric Manifold Warping and Weight Recombination

By applying the SpFI mapping, the influence of each point on the posterior state is dynamically adjusted. Geometrically, this is equivalent to mapping the sigma points onto an iteration-specific **Fuzzy Manifold** where the radius of influence in any given direction is a function of the local sensor agreement ϕ_i.

To maintain the unbiased property of the filter, the nominal weights W_i are corrected to produce the fuzzy-adapted weights W_i^ϕ:

$$W_i^\phi = \frac{W_i \cdot \phi_i}{\sum_{j=0}^{2L}(W_j \cdot \phi_j)} \tag{6}$$

This allows the filter to selectively ignore segments of the uncertainty ellipsoid that have drifted into non-physical or saturated sensor regions without discarding the valid information present in orthogonal dimensions. The resulting posterior mean $\hat{x}_k$ is calculated as:

$$\hat{x}_k = \sum_{i=0}^{2L} W_i^\phi \cdot \mathcal{X}_{i,k|k-1} \tag{7}$$

3 Trust Membership Function Design

The performance of the Sigma Point Fuzzy Inference (SpFI) system depends on the calibration of the trust membership function $\mu_{\text{trust}}(\zeta_i)$. Because the whitened innovation mapping ζ_i is a sum of squared Gaussian variables, it theoretically follows a χ^2 distribution with m degrees of freedom, where m is the dimension of the measurement vector $\boldsymbol{z}_k$.

3.1 Statistical Manifold Boundaries

To ensure the Fuzzy Adaptive Manifold only deforms when statistically necessary, we define two critical thresholds, Γ_{low} and Γ_{high}, derived from the χ^2 Cumulative Distribution Function (CDF):

- **High-Trust Threshold (Γ_{low}):** The boundary within which a sigma point is considered a nominal "inlier." Typically set at the 95% confidence interval (e.g., $\Gamma_{low} \approx 7.81$ for a 3-axis accelerometer).
- **Outlier Threshold (Γ_{high}):** The boundary beyond which a sigma point is considered non-physical or corrupted.

3.2 Trapezoidal SpFI Mapping

To maintain deterministic execution on the STM32F401, we implement a **Trapezoidal Membership Function** for the trust factor ϕ_i. This provides a smooth, linear transition between full trust and the manifold floor $\phi_{\min}$:

$$\mathrm{FIS}(\zeta_i) = \begin{cases} 1.0 & \zeta_i \leq \Gamma_{low} \\ \frac{\Gamma_{high} - \zeta_i}{\Gamma_{high} - \Gamma_{low}} & \Gamma_{low} < \zeta_i < \Gamma_{high} \\ 0.0 & \zeta_i \geq \Gamma_{high} \end{cases} \tag{8}$$

The final importance factor ϕ_i is then constrained by the safety guard:

$$\phi_i = \max\left(\phi_{\min}, \mathrm{FIS}(\zeta_i)\right) \tag{9}$$

3.3 Geometric Interpretation of the Trust Curve

As ζ_i increases beyond Γ_{low}, the SpFI mapping begins to "pinch" the uncertainty manifold in that specific direction. While the sigma point $\mathcal{X}_i$ remains at its canonical distance from the mean, its influence on the posterior estimate is attenuated. At $\zeta_i \geq \Gamma_{high}$, the point is effectively "quarantined," and its contribution to the state update is reduced to the minimal floor $\phi_{\min}$. This allows the filter to remain responsive to valid maneuvers while becoming strictly robust against directional sensor anomalies.

4 Dual-Stage FAM Integration

The Fuzzy Adaptive Manifold is applied at both the *Predict* and *Update* phases of the SR-UKF to ensure end-to-end filter integrity.

4.1 Predictive Manifold (Kinematic Constraint)

During the state prediction phase, $\hat{x}_k^- = f(\hat{x}_{k-1}, \boldsymbol{u}_k)$, the FAM monitors the kinematic feasibility of each sigma point trajectory. If a specific point $\mathcal{X}_{i,k|k-1}$ violates structural constraints (e.g., unit quaternion divergence), the SpFI reduces its contribution to the predicted mean. This ensures that the *a priori* estimate remains within the physically valid manifold before measurements are integrated.

4.2 Corrective Manifold (Measurement Validation)

In the update phase, the FAM utilizes the innovation whitening mapping ζ_i. As the Accelerometer data $\boldsymbol{z}_{accel}$ is ingested, the SpFI evaluates the distance between the sensor reality and the manifold's predicted observations $\mathcal{Y}_{i,k}$.

$$\hat{x}_k = \sum_{i=0}^{2L} W_i^{\phi} \cdot \mathcal{X}_{i,k|k-1} \tag{10}$$

By scaling the measurement noise R_s in conjunction with the FAM weights, the filter achieves a "surgical" snap to the gravity vector without the jitter typically associated with low R values.

4.3 Mathematical Notation

Throughout this paper, we adopt the standard Kalman filtering notation to distinguish between the predictive and corrective phases of the estimation cycle:

- $\hat{\boldsymbol{x}}_k^-$ denotes the *a priori* state estimate at time k. This represents the "predicted" state derived solely from the kinematic model and the previous state $\hat{\boldsymbol{x}}_{k-1}$, prior to incorporating the current sensor measurement.
- $\hat{\boldsymbol{x}}_k$ (or $\hat{\boldsymbol{x}}_k^+$) denotes the *a posteriori* state estimate. This represents the "corrected" state after the Sigma Point Fuzzy Inference (SpFI) and Kalman Gain have been applied to the measurement $\boldsymbol{z}_k$.
- Similarly, $\mathbf{S}_k^-$ and $\mathbf{S}_k$ represent the square-root covariance matrices before and after the measurement update, respectively.

5 Hierarchical Innovation Mapping

A critical distinction in the Fuzzy Adaptive Manifold (FAM) architecture is the separation of point-wise validation from the global state update. To achieve directional robustness, the SpFI mechanism operates on two distinct innovation vectors: the *Point-wise Innovation* (ζ_i) and the *Posterior Mean Innovation* ($\boldsymbol{\nu}$).

5.1 Stage I: Local Manifold Shaping (Point-wise)

Before the filter mean $\hat{z}$ is calculated, each individual sigma point trajectory $\mathcal{X}_i$ is evaluated against the raw measurement $\boldsymbol{z}_k$. We define the point-wise innovation $\boldsymbol{\epsilon}_i$ as the discrepancy between the sensor reality and the i-th predicted trajectory:

$$\boldsymbol{\epsilon}_i = \boldsymbol{z}_k - h(\mathcal{X}_{i,k|k-1}) \tag{11}$$

The scalar mapping $\zeta_i = \boldsymbol{\epsilon}_i^T R^{-1} \boldsymbol{\epsilon}_i$ is then used as the input to the SpFI system. This stage identifies which points of the uncertainty ellipsoid are inconsistent with the incoming data. By calculating ζ_i for all $2L+1$ points, the FAM synthesizes the trust factors ϕ_i required to warp the manifold.

5.2 Stage II: Global Mean Integration (Posterior)

Once the manifold has been warped via the fuzzy weights W_i^ϕ, the *Fuzzy Predicted Mean* $\hat{z}_\phi$ is synthesized:

$$\hat{z}_\phi = \sum_{i=0}^{2L} W_i^\phi \mathcal{Y}_{i,k} \tag{12}$$

The final filter innovation $\boldsymbol{\nu}$—used for the Kalman gain update and the IMM likelihood calculation—is then derived relative to this fuzzy mean:

$$\boldsymbol{\nu} = \boldsymbol{z}_k - \hat{z}_\phi \tag{13}$$

5.3 Mathematical Advantage of Dual-Innovation

This hierarchical approach prevents the **Mean-Shift Bias** typically found in standard UKFs. In a standard filter, a single outlier sigma point pulls the mean $\hat{z}$ toward the outlier, resulting in a corrupted innovation $\boldsymbol{\nu}$. Under FAM, the SpFI "quarantines" the outlier at the point-wise stage (Stage I). Consequently, the fuzzy mean $\hat{z}_\phi$ remains centered on the reliable data, ensuring that the final innovation $\boldsymbol{\nu}$ accurately reflects the true state error rather than the sensor's transient noise.

6 Predictive Manifold Expansion

The FAM architecture distinguishes between *volume expansion* (Predict) and *geometric warping* (Update). During the prediction phase, the manifold undergoes a uniform scaling based on the previous iteration's health metric ($\epsilon_{nis,k-1}$). This ensures that the sigma points $\mathcal{X}_i$ encompass a sufficiently large volume of the state-space to capture high-dynamic transients:

$$\mathbf{Q}_{s,k} = \mathrm{FIS}_{vol}(\epsilon_{nis,k-1}) \cdot \mathbf{Q}_{s,base} \tag{14}$$

By expanding the manifold volume *a priori*, the subsequent update stage is better conditioned to identify and reject directional outliers through SpFI warping.

7 Metrics for Manifold Mapping: Mahalanobis vs. Euclidean

The efficacy of the Sigma Point Fuzzy Inference (SpFI) depends on the distance metric used to map the sigma point discrepancy into the scalar trust factor ϕ_i. We analyze two candidates: the theoretically optimal *Mahalanobis Mapping* and the computationally efficient *Euclidean Proxy*.

7.1 The Superiority of Mahalanobis Whitening

The optimal mapping for a Gaussian uncertainty manifold is the Mahalanobis distance, which accounts for the cross-correlation between measurement dimensions. Given the residual square-root covariance $S_{r,k}$, the Mahalanobis mapping ζ_i^M is:

$$\zeta_i^M = \|S_{r,k}^{-1}(\boldsymbol{z}_k - \mathcal{Y}_{i,k})\|^2 \tag{15}$$

Proof of Precision: The Mahalanobis metric is the only mapping that satisfies *affine invariance*. Because $S_{r,k}$ encodes the principal axes of the uncertainty ellipsoid, ζ_i^M effectively "whitens" the innovation, transforming the

ellipsoid into a perfect unit hypersphere. Mathematically, if $\boldsymbol{\nu} \sim \mathcal{N}(0, P)$, then the Mahalanobis mapping ensures that:

$$E[\zeta_i^M] = m \quad \text{and} \quad \text{Var}[\zeta_i^M] = 2m \tag{16}$$

where m is the measurement dimension. This allows for a universal FIS threshold (Γ) that is independent of the sensor's current noise levels or axis-specific scaling.

7.2 The Euclidean Proxy and Computational Complexity

In high-frequency embedded systems (> 200Hz), calculating ζ_i^M for all $2L + 1$ sigma points requires $2L + 1$ back-substitution operations ($O(L \cdot m^2)$). For a 7D state and 3D measurement, this adds significant latency. We define the *Euclidean Proxy* ζ_i^E:

$$\zeta_i^E = \frac{1}{R_{\text{scalar}}^2} \|\boldsymbol{z}_k - \mathcal{Y}_{i,k}\|^2 \tag{17}$$

7.3 Selection Rationale for FAM

We implement the Euclidean Proxy for the SpFI mapping for the following reasons:

- **Causality:** At the start of the update loop, the current $S_{r,k}$ is unknown. Using the $S_{r,k-1}$ from the previous frame introduces a temporal lag in the manifold warping.
- **Directional Sensitivity:** While the Euclidean metric ignores cross-correlation, it remains highly sensitive to axis-specific outliers (e.g., sensor clipping), which is the primary failure mode the FAM is designed to mitigate.
- **Numerical Stability:** The Euclidean metric is inherently decoupled from the state covariance, preventing a cross-correlation where a corrupted S matrix could falsely validate an outlier sigma point.

7.4 Innovation and Mapping Nomenclature

To maintain mathematical and computational clarity within the FAM framework, we distinguish between two hierarchical levels of innovation:

- **Point-wise Innovation Mapping (ζ_i):** A set of $2L+1$ scalar values ($i = 0, \ldots, 14$) representing the Mahalanobis or Euclidean distance between the i-th predicted sigma point trajectory and the raw sensor measurement. These scalars are the primary inputs to the SpFI system and are used to warp the *a priori* manifold.

- **Mean Innovation Vector ($\boldsymbol{\nu}$):** A single m-dimensional vector (e.g., 3×1 for an accelerometer) representing the residual between the sensor measurement and the *Fuzzy Predicted Mean* ($\hat{z}_\phi$). This vector is used for the final state correction and the IMM likelihood calculation.

In the embedded implementation, the array `ctx->zeta[SIGMA_COUNT]` stores the point-wise mapping values for diagnostic visualization, while the vector `ctx->innovation_mean` stores the residual used for the Kalman gain multiplication.

7.5 Statistical Weighting vs. Geometric Warping

While the FAM architecture utilizes **Importance Weighting** to attenuate outlier influence, it effectively performs a **Pseudo-Warping** of the uncertainty manifold. Mathematically, de-weighting a sigma point along a specific principal axis is dual to contracting the manifold's radius in that dimension.

By maintaining the canonical geometric locations of the sigma points while dynamically adjusting their weights W_i^ϕ, we achieve the robustness of a warped manifold without the $O(L^2)$ computational cost of re-generating the point set or the risk of inducing numerical singularity in the Cholesky factor S.

8 Conclusion

This paper has presented a novel framework for robust embedded state estimation: **Fuzzy Adaptive Manifold (FAM)**. By transitioning from global covariance scaling to point-specific **Sigma Point Fuzzy Inference (SpFI)**, we have demonstrated a method for warping the uncertainty distribution in response to directional sensor anomalies.

The proposed architecture achieves three critical objectives for high-performance IMU fusion on constrained hardware like the STM32F401:

1. **Directional Robustness:** The SpFI mechanism identifies and *quarantines* outliers at the sigma-point level, allowing the filter to maintain precision in reliable dimensions even during localized sensor saturation.

2. **Geometric Adaptivity:** Through the use of iteration-specific manifolds, the filter effectively transforms the rigid Unscented Transform into a deformable trust-volume that adapts to the real-time innovation environment.

3. **Computational Determinism:** By utilizing a *Euclidean Proxy* for the manifold mapping and a Zero-Order Sugeno FIS, the FAM logic introduces negligible $O(L)$ overhead, preserving the 800Hz update rate required for flight control.

9 Benchmark Performance

The FAM-SR-UKF was benchmarked on an ARM Cortex-M4 (STM32F401) running at 84 MHz. Utilizing the CMSIS-DSP library and the SpFI Euclidean Proxy, the total execution time for a 7-dimensional state update was recorded at 1.25 ms. This corresponds to an operational frequency of 800 Hz, providing a $4\times$ margin over the required 200 Hz flight control loop. This performance demonstrates that Fuzzy Adaptive Manifolds provide high-order robustness without the traditional computational penalties of non-parametric estimators.

Future work for Jitterware, Inc. will focus on the integration of FAM with Interacting Multiple Model (IMM) likelihoods to create a multi-tiered defense against both structural maneuvers and sensor-level failures. The result is a filter that behaves with the statistical elegance of a Particle Filter but operates with the numerical efficiency of a Square-Root Kalman Filter.

9.1 Visual Verification of Manifold Warping

To verify the real-time deformation of the trust manifold, the point-wise innovation metrics ζ_i were streamed to a 3D visualization engine (Unity). By mapping ζ_i to the visual scale and chromaticity of the sigma point primitives (ϕ_i), we confirmed that the FAM correctly identifies and attenuates directional outliers. During simulated 2g impact transients, the manifold was observed to asymmetrically contract along the axis of impact, preserving the integrity of the orthogonal state estimates.

10 Acknowledgements

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